

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI)

DEEP LEARNING LAB RECORD

LAB - 02

Student Details

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Course Name: Deep Learning

Course Code: 22AIE304

Experiment Title

Multilayer Perceptron (MLP) and Hyperparameter Tuning using the Iris Dataset

Aim

To implement a Multilayer Perceptron (MLP) using TensorFlow/Keras and analyze the impact of different activation functions and learning rates on model performance.

Objective

- Implement an MLP for multiclass classification.
 - Understand Forward and Backward Propagation.
 - Compare Sigmoid and ReLU activation functions.
 - Compare different learning rates.
 - Evaluate model accuracy using the Iris dataset.
-

Dataset

Dataset Name : Iris Dataset

Number of Samples : 150

Features :

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Classes :

- Iris Setosa
 - Iris Versicolor
 - Iris Virginica
-

Software Requirements

- Python 3.x
 - TensorFlow
 - NumPy
 - Pandas
 - Matplotlib
 - Scikit-Learn
 - Jupyter Notebook / Anaconda
-

Required Libraries

- tensorflow
- numpy
- pandas
- matplotlib
- sklearn

```
In [1]: !pip install tensorflow numpy pandas matplotlib scikit-learn
```

Requirement already satisfied: tensorflow in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.21.0)

Requirement already satisfied: numpy in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.2.6)

Requirement already satisfied: pandas in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (2.3.3)

Requirement already satisfied: matplotlib in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (3.10.9)

Requirement already satisfied: scikit-learn in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (1.7.2)

Requirement already satisfied: libclang>=13.0.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (18.1.1)

Requirement already satisfied: flatbuffers>=25.9.23 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (25.12.19)

Requirement already satisfied: google_pasta>=0.1.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.2.0)

Requirement already satisfied: six>=1.12.0 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (1.17.0)

Requirement already satisfied: setuptools in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (65.5.0)

Requirement already satisfied: ml_dtypes<1.0.0,>=0.5.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.5.4)

Requirement already satisfied: astunparse>=1.6.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (1.6.3)

Requirement already satisfied: keras>=3.12.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.12.3)

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Requirement already satisfied: h5py<3.15.0,>=3.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (3.14.0)

Requirement already satisfied: typing_extensions>=3.6.6 in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (4.15.0)

Requirement already satisfied: packaging in c:\users\msi\appdata\roaming\python\python310\site-packages (from tensorflow) (25.0)

Requirement already satisfied: wrapt>=1.11.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.2.2)

Requirement already satisfied: requests<3,>=2.21.0 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (2.34.2)

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Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from tensorflow) (0.7.0)

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Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\msi\appdata\roaming\python\python310\site-packages (from pandas) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from pandas) (2026.2)

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Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.5.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\msi\appdata\local\programs\python\python310\lib\site-packages (from matplotlib) (1.3.2)

Requirement already satisfied: cycycler>=0.10 in c:\users\msi\appdata\local\program

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Requirement already satisfied: pyparsing>=3 in c:\users\msi\appdata\local\program
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Requirement already satisfied: pillow>=8 in c:\users\msi\appdata\local\programs\p
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Requirement already satisfied: joblib>=1.2.0 in c:\users\msi\appdata\local\progra
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Requirement already satisfied: threadpoolctl>=3.1.0 in c:\users\msi\appdata\local
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Requirement already satisfied: scipy>=1.8.0 in c:\users\msi\appdata\local\program
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Requirement already satisfied: wheel<1.0,>=0.23.0 in c:\users\msi\appdata\local\p
rograms\python\python310\lib\site-packages (from astunparse>=1.6.0->tensorflow)
(0.47.0)
Requirement already satisfied: rich in c:\users\msi\appdata\local\programs\python
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Requirement already satisfied: namex in c:\users\msi\appdata\local\programs\pytho
n\python310\lib\site-packages (from keras>=3.12.0->tensorflow) (0.1.0)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\users\msi\appdata\l
ocal\programs\python\python310\lib\site-packages (from requests<3,>=2.21.0->tenso
rflow) (3.4.7)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\msi\appdata\local\pro
grams\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2.7.0)
Requirement already satisfied: idna<4,>=2.5 in c:\users\msi\appdata\local\program
s\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow) (3.1
6)
Requirement already satisfied: certifi>=2023.5.7 in c:\users\msi\appdata\local\pr
ograms\python\python310\lib\site-packages (from requests<3,>=2.21.0->tensorflow)
(2026.5.20)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\msi\appdata\loca
l\programs\python\python310\lib\site-packages (from rich->keras>=3.12.0->tensorfl
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in c:\users\msi\appdata\ro
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9.2)
Requirement already satisfied: mdurl~=0.1 in c:\users\msi\appdata\local\programs
\python\python310\lib\site-packages (from markdown-it-py>=2.2.0->rich->keras>=3.1
2.0->tensorflow) (0.1.2)

[notice] A new release of pip is available: 23.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

In [2]: *# Import Required Libraries*

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

import tensorflow as tf
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam
```

In [3]: `import pandas as pd`

```
columns = [
    "SepalLength",
    "SepalWidth",
    "PetalLength",
    "PetalWidth",
    "Species"
]

dataset = pd.read_csv(
    "iris.data",
    header=None,
    names=columns
)

dataset.head()
```

Out[3]:

	SepalLength	SepalWidth	PetalLength	PetalWidth	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [4]: `# Dataset Information`

```
dataset.info()

dataset.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   SepalLength     150 non-null   float64
1   SepalWidth      150 non-null   float64
2   PetalLength     150 non-null   float64
3   PetalWidth      150 non-null   float64
4   Species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

Out[4]:

	SepalLength	SepalWidth	PetalLength	PetalWidth
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [5]: # Preprocessing

# Features
X = dataset.iloc[:, :-1].values

# Target Labels
y = dataset.iloc[:, -1].values

# Encode Species Labels into numeric values
encoder = LabelEncoder()
y = encoder.fit_transform(y)

print("Features Shape:", X.shape)
print("Labels Shape:", y.shape)
print("Encoded Classes:", encoder.classes_)
```

```
Features Shape: (150, 4)
Labels Shape: (150,)
Encoded Classes: ['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']
```

```
In [6]: # Feature Scaling

scaler = StandardScaler()

X = scaler.fit_transform(X)
```

```
In [7]: # Train Test Split

X_train, X_test, y_train, y_test = train_test_split(

    X,

    y,

    test_size=0.20,

    random_state=42

)

print(X_train.shape)
```

```
print(X_test.shape)
```

```
(120, 4)
(30, 4)
```

In [8]: *# Build the Multilayer Perceptron (MLP)*

```
model = Sequential()

model.add(Dense(16, activation='relu', input_shape=(4,)))

model.add(Dense(8, activation='relu'))

model.add(Dense(3, activation='softmax'))

model.summary()
```

c:\Users\msi\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\core\dense.py:95: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 16)	80
dense_1 (Dense)	(None, 8)	136
dense_2 (Dense)	(None, 3)	27

Total params: 243 (972.00 B)

Trainable params: 243 (972.00 B)

Non-trainable params: 0 (0.00 B)

In [9]: *# Compile the Model*

```
model.compile(
    optimizer=Adam(learning_rate=0.01),
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

In [10]: *# Train the Model*

```
history = model.fit(
    X_train,
    y_train,
    epochs=100,
    batch_size=16,
    validation_split=0.2,
```

```
verbose=1
```

```
)
```


Epoch 1/100
6/6  1s 29ms/step - accuracy: 0.5521 - loss: 0.9857 - val_accuracy: 0.7917 - val_loss: 0.7010
Epoch 2/100
6/6  0s 11ms/step - accuracy: 0.8438 - loss: 0.6497 - val_accuracy: 0.8750 - val_loss: 0.5182
Epoch 3/100
6/6  0s 11ms/step - accuracy: 0.8438 - loss: 0.4537 - val_accuracy: 0.9167 - val_loss: 0.4036
Epoch 4/100
6/6  0s 10ms/step - accuracy: 0.8646 - loss: 0.3478 - val_accuracy: 0.9167 - val_loss: 0.3223
Epoch 5/100
6/6  0s 11ms/step - accuracy: 0.9167 - loss: 0.2746 - val_accuracy: 0.9167 - val_loss: 0.2742
Epoch 6/100
6/6  0s 11ms/step - accuracy: 0.9271 - loss: 0.2288 - val_accuracy: 0.9583 - val_loss: 0.2274
Epoch 7/100
6/6  0s 11ms/step - accuracy: 0.9375 - loss: 0.1905 - val_accuracy: 0.9583 - val_loss: 0.1832
Epoch 8/100
6/6  0s 11ms/step - accuracy: 0.9375 - loss: 0.1604 - val_accuracy: 0.9583 - val_loss: 0.1779
Epoch 9/100
6/6  0s 11ms/step - accuracy: 0.9583 - loss: 0.1402 - val_accuracy: 0.9583 - val_loss: 0.1379
Epoch 10/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.1231 - val_accuracy: 0.9583 - val_loss: 0.1477
Epoch 11/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.1098 - val_accuracy: 0.9583 - val_loss: 0.1238
Epoch 12/100
6/6  0s 12ms/step - accuracy: 0.9688 - loss: 0.1010 - val_accuracy: 0.9583 - val_loss: 0.1157
Epoch 13/100
6/6  0s 12ms/step - accuracy: 0.9583 - loss: 0.0936 - val_accuracy: 0.9583 - val_loss: 0.1200
Epoch 14/100
6/6  0s 12ms/step - accuracy: 0.9583 - loss: 0.0874 - val_accuracy: 0.9583 - val_loss: 0.1314
Epoch 15/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0830 - val_accuracy: 0.9583 - val_loss: 0.0966
Epoch 16/100
6/6  0s 11ms/step - accuracy: 0.9688 - loss: 0.0886 - val_accuracy: 0.9583 - val_loss: 0.0717
Epoch 17/100
6/6  0s 9ms/step - accuracy: 0.9688 - loss: 0.0737 - val_accuracy: 0.9583 - val_loss: 0.1172
Epoch 18/100
6/6  0s 11ms/step - accuracy: 0.9688 - loss: 0.0816 - val_accuracy: 0.9583 - val_loss: 0.1072
Epoch 19/100
6/6  0s 11ms/step - accuracy: 0.9583 - loss: 0.0791 - val_accuracy: 0.9583 - val_loss: 0.0586
Epoch 20/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0844 - val_accuracy: 0.9583 - val_loss: 0.1190

Epoch 21/100
6/6  0s 11ms/step - accuracy: 0.9583 - loss: 0.0733 - val_accuracy: 0.9583 - val_loss: 0.1045
Epoch 22/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0671 - val_accuracy: 0.9583 - val_loss: 0.0975
Epoch 23/100
6/6  0s 14ms/step - accuracy: 0.9583 - loss: 0.0654 - val_accuracy: 0.9583 - val_loss: 0.0650
Epoch 24/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.0696 - val_accuracy: 0.9583 - val_loss: 0.0945
Epoch 25/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0657 - val_accuracy: 0.9583 - val_loss: 0.0771
Epoch 26/100
6/6  0s 9ms/step - accuracy: 0.9583 - loss: 0.0602 - val_accuracy: 0.9583 - val_loss: 0.0563
Epoch 27/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.0650 - val_accuracy: 0.9583 - val_loss: 0.0854
Epoch 28/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0582 - val_accuracy: 0.9583 - val_loss: 0.0754
Epoch 29/100
6/6  0s 11ms/step - accuracy: 0.9583 - loss: 0.0633 - val_accuracy: 0.9583 - val_loss: 0.0701
Epoch 30/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.0843 - val_accuracy: 0.9167 - val_loss: 0.1184
Epoch 31/100
6/6  0s 11ms/step - accuracy: 0.9583 - loss: 0.0635 - val_accuracy: 1.0000 - val_loss: 0.0366
Epoch 32/100
6/6  0s 10ms/step - accuracy: 0.9583 - loss: 0.0638 - val_accuracy: 0.9583 - val_loss: 0.0648
Epoch 33/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0560 - val_accuracy: 0.9583 - val_loss: 0.0843
Epoch 34/100
6/6  0s 9ms/step - accuracy: 0.9792 - loss: 0.0570 - val_accuracy: 0.9583 - val_loss: 0.0921
Epoch 35/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0546 - val_accuracy: 0.9583 - val_loss: 0.0769
Epoch 36/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0570 - val_accuracy: 0.9583 - val_loss: 0.0898
Epoch 37/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0558 - val_accuracy: 0.9583 - val_loss: 0.0686
Epoch 38/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0524 - val_accuracy: 0.9583 - val_loss: 0.0806
Epoch 39/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0518 - val_accuracy: 0.9583 - val_loss: 0.0945
Epoch 40/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0546 - val_accuracy: 0.9583 - val_loss: 0.0660

Epoch 41/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0560 - val_accuracy: 0.9583 - val_loss: 0.1095
Epoch 42/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0549 - val_accuracy: 0.9583 - val_loss: 0.0555
Epoch 43/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0554 - val_accuracy: 0.9583 - val_loss: 0.0703
Epoch 44/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0500 - val_accuracy: 0.9583 - val_loss: 0.0805
Epoch 45/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0536 - val_accuracy: 0.9583 - val_loss: 0.1111
Epoch 46/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0487 - val_accuracy: 0.9583 - val_loss: 0.0672
Epoch 47/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0517 - val_accuracy: 0.9583 - val_loss: 0.0924
Epoch 48/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0501 - val_accuracy: 0.9583 - val_loss: 0.0932
Epoch 49/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0482 - val_accuracy: 0.9583 - val_loss: 0.0944
Epoch 50/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0473 - val_accuracy: 0.9583 - val_loss: 0.0889
Epoch 51/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0458 - val_accuracy: 0.9583 - val_loss: 0.0876
Epoch 52/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0467 - val_accuracy: 0.9583 - val_loss: 0.0909
Epoch 53/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0446 - val_accuracy: 0.9583 - val_loss: 0.1377
Epoch 54/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0444 - val_accuracy: 0.9583 - val_loss: 0.0962
Epoch 55/100
6/6  0s 9ms/step - accuracy: 0.9792 - loss: 0.0431 - val_accuracy: 0.9583 - val_loss: 0.0730
Epoch 56/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0442 - val_accuracy: 0.9583 - val_loss: 0.0738
Epoch 57/100
6/6  0s 9ms/step - accuracy: 0.9792 - loss: 0.0467 - val_accuracy: 0.9583 - val_loss: 0.1336
Epoch 58/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0470 - val_accuracy: 0.9583 - val_loss: 0.1081
Epoch 59/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0366 - val_accuracy: 0.9583 - val_loss: 0.0464
Epoch 60/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0601 - val_accuracy: 0.9583 - val_loss: 0.0569

Epoch 61/100
6/6  0s 9ms/step - accuracy: 0.9792 - loss: 0.0427 - val_accuracy: 0.9167 - val_loss: 0.1541
Epoch 62/100
6/6  0s 11ms/step - accuracy: 0.9688 - loss: 0.0539 - val_accuracy: 0.9583 - val_loss: 0.0475
Epoch 63/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0510 - val_accuracy: 0.9583 - val_loss: 0.0484
Epoch 64/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0529 - val_accuracy: 0.9583 - val_loss: 0.1032
Epoch 65/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0399 - val_accuracy: 0.9583 - val_loss: 0.0656
Epoch 66/100
6/6  0s 11ms/step - accuracy: 0.9896 - loss: 0.0378 - val_accuracy: 0.9583 - val_loss: 0.0577
Epoch 67/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0433 - val_accuracy: 0.9583 - val_loss: 0.0513
Epoch 68/100
6/6  0s 13ms/step - accuracy: 0.9792 - loss: 0.0429 - val_accuracy: 0.9583 - val_loss: 0.1101
Epoch 69/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0370 - val_accuracy: 0.9583 - val_loss: 0.0757
Epoch 70/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0492 - val_accuracy: 1.0000 - val_loss: 0.0361
Epoch 71/100
6/6  0s 10ms/step - accuracy: 0.9688 - loss: 0.0404 - val_accuracy: 0.9167 - val_loss: 0.1221
Epoch 72/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0396 - val_accuracy: 0.9583 - val_loss: 0.0683
Epoch 73/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0355 - val_accuracy: 0.9583 - val_loss: 0.0554
Epoch 74/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0356 - val_accuracy: 0.9583 - val_loss: 0.0601
Epoch 75/100
6/6  0s 11ms/step - accuracy: 0.9896 - loss: 0.0319 - val_accuracy: 0.9583 - val_loss: 0.0967
Epoch 76/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0339 - val_accuracy: 0.9583 - val_loss: 0.0776
Epoch 77/100
6/6  0s 13ms/step - accuracy: 0.9792 - loss: 0.0343 - val_accuracy: 0.9583 - val_loss: 0.0857
Epoch 78/100
6/6  0s 12ms/step - accuracy: 0.9792 - loss: 0.0311 - val_accuracy: 0.9583 - val_loss: 0.0692
Epoch 79/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0324 - val_accuracy: 0.9583 - val_loss: 0.0580
Epoch 80/100
6/6  0s 11ms/step - accuracy: 0.9896 - loss: 0.0367 - val_accuracy: 0.9583 - val_loss: 0.0497

Epoch 81/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0469 - val_accuracy: 0.9583 - val_loss: 0.0752
Epoch 82/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0318 - val_accuracy: 1.0000 - val_loss: 0.0217
Epoch 83/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0354 - val_accuracy: 0.9583 - val_loss: 0.0542
Epoch 84/100
6/6  0s 11ms/step - accuracy: 0.9792 - loss: 0.0291 - val_accuracy: 0.9583 - val_loss: 0.0762
Epoch 85/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0310 - val_accuracy: 1.0000 - val_loss: 0.0390
Epoch 86/100
6/6  0s 10ms/step - accuracy: 0.9792 - loss: 0.0409 - val_accuracy: 1.0000 - val_loss: 0.0199
Epoch 87/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0252 - val_accuracy: 0.9167 - val_loss: 0.0996
Epoch 88/100
6/6  0s 9ms/step - accuracy: 0.9792 - loss: 0.0431 - val_accuracy: 1.0000 - val_loss: 0.0314
Epoch 89/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0296 - val_accuracy: 0.9583 - val_loss: 0.0466
Epoch 90/100
6/6  0s 12ms/step - accuracy: 0.9896 - loss: 0.0258 - val_accuracy: 1.0000 - val_loss: 0.0477
Epoch 91/100
6/6  0s 11ms/step - accuracy: 0.9896 - loss: 0.0326 - val_accuracy: 1.0000 - val_loss: 0.0388
Epoch 92/100
6/6  0s 11ms/step - accuracy: 1.0000 - loss: 0.0255 - val_accuracy: 1.0000 - val_loss: 0.0564
Epoch 93/100
6/6  0s 11ms/step - accuracy: 1.0000 - loss: 0.0251 - val_accuracy: 1.0000 - val_loss: 0.0593
Epoch 94/100
6/6  0s 10ms/step - accuracy: 0.9896 - loss: 0.0246 - val_accuracy: 1.0000 - val_loss: 0.0435
Epoch 95/100
6/6  0s 9ms/step - accuracy: 0.9896 - loss: 0.0258 - val_accuracy: 1.0000 - val_loss: 0.0350
Epoch 96/100
6/6  0s 32ms/step - accuracy: 1.0000 - loss: 0.0229 - val_accuracy: 0.9167 - val_loss: 0.0681
Epoch 97/100
6/6  0s 39ms/step - accuracy: 1.0000 - loss: 0.0222 - val_accuracy: 1.0000 - val_loss: 0.0225
Epoch 98/100
6/6  0s 38ms/step - accuracy: 0.9896 - loss: 0.0338 - val_accuracy: 0.9583 - val_loss: 0.0535
Epoch 99/100
6/6  0s 35ms/step - accuracy: 1.0000 - loss: 0.0227 - val_accuracy: 0.9583 - val_loss: 0.0538
Epoch 100/100
6/6  0s 39ms/step - accuracy: 1.0000 - loss: 0.0201 - val_accuracy: 1.0000 - val_loss: 0.0390

In [11]: *# Evaluate the Model*

```
loss, accuracy = model.evaluate(X_test, y_test)

print("Test Loss :", loss)

print("Test Accuracy :", accuracy)
```

1/1 ————— 0s 24ms/step - accuracy: 0.9667 - loss: 0.0350
Test Loss : 0.03500407189130783
Test Accuracy : 0.9666666388511658

In [12]: *# Predict*

```
predictions = model.predict(X_test)

predicted_classes = np.argmax(predictions, axis=1)

print(predicted_classes[:10])
```

1/1 ————— 0s 37ms/step
[1 0 2 1 1 0 1 2 2 1]

In [13]: *# Compare Actual and Predicted Values*

```
comparison = pd.DataFrame({
    "Actual": y_test,
    "Predicted": predicted_classes
})

comparison.head(10)
```

Out[13]:

	Actual	Predicted
--	--------	-----------

0	1	1
1	0	0
2	2	2
3	1	1
4	1	1
5	0	0
6	1	1
7	2	2
8	1	2
9	1	1

In [14]: **from** sklearn.metrics **import** confusion_matrix

```
cm = confusion_matrix(y_test, predicted_classes)

print(cm)
```

```
[[10  0  0]
 [ 0  8  1]
 [ 0  0 11]]
```

```
In [15]: from sklearn.metrics import classification_report

print(classification_report(y_test, predicted_classes))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	0.89	0.94	9
2	0.92	1.00	0.96	11
accuracy			0.97	30
macro avg	0.97	0.96	0.97	30
weighted avg	0.97	0.97	0.97	30

```
In [16]: plt.figure(figsize=(8,5))

plt.plot(history.history['accuracy'], label='Training Accuracy')

plt.plot(history.history['val_accuracy'], label='Validation Accuracy')

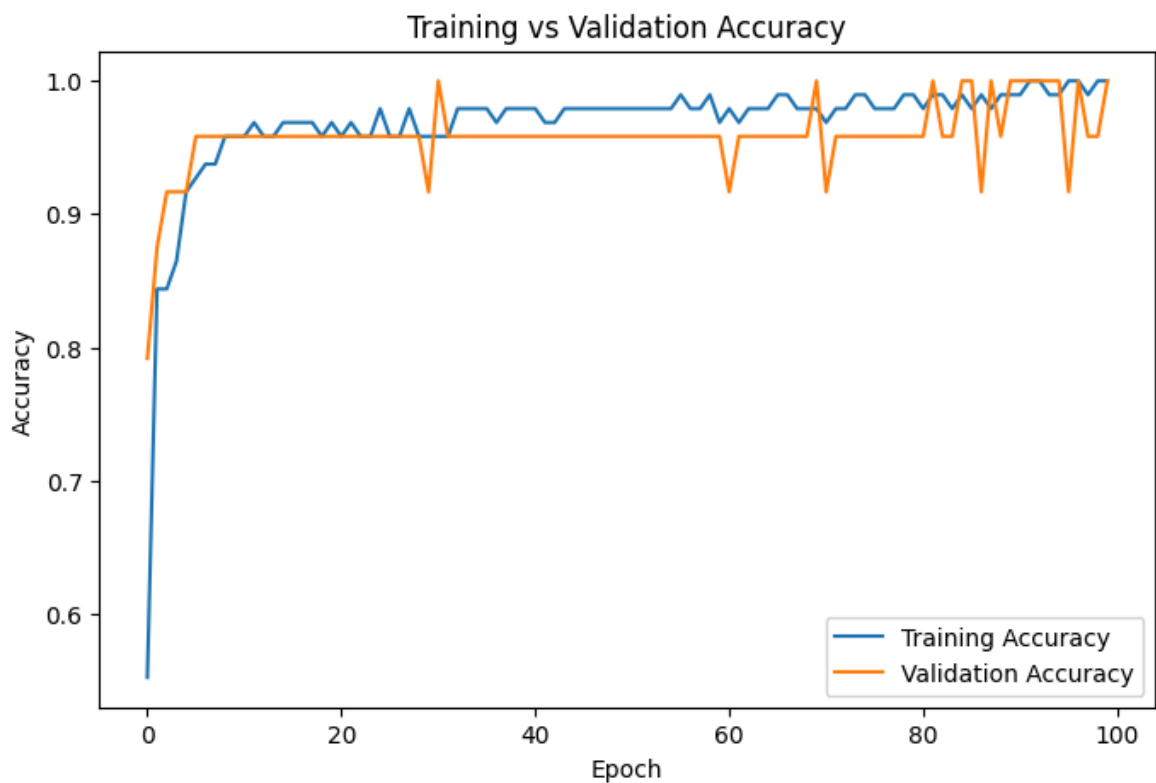
plt.xlabel("Epoch")

plt.ylabel("Accuracy")

plt.title("Training vs Validation Accuracy")

plt.legend()

plt.show()
```



```
In [17]: plt.figure(figsize=(8,5))

plt.plot(history.history['loss'], label='Training Loss')

plt.plot(history.history['val_loss'], label='Validation Loss')

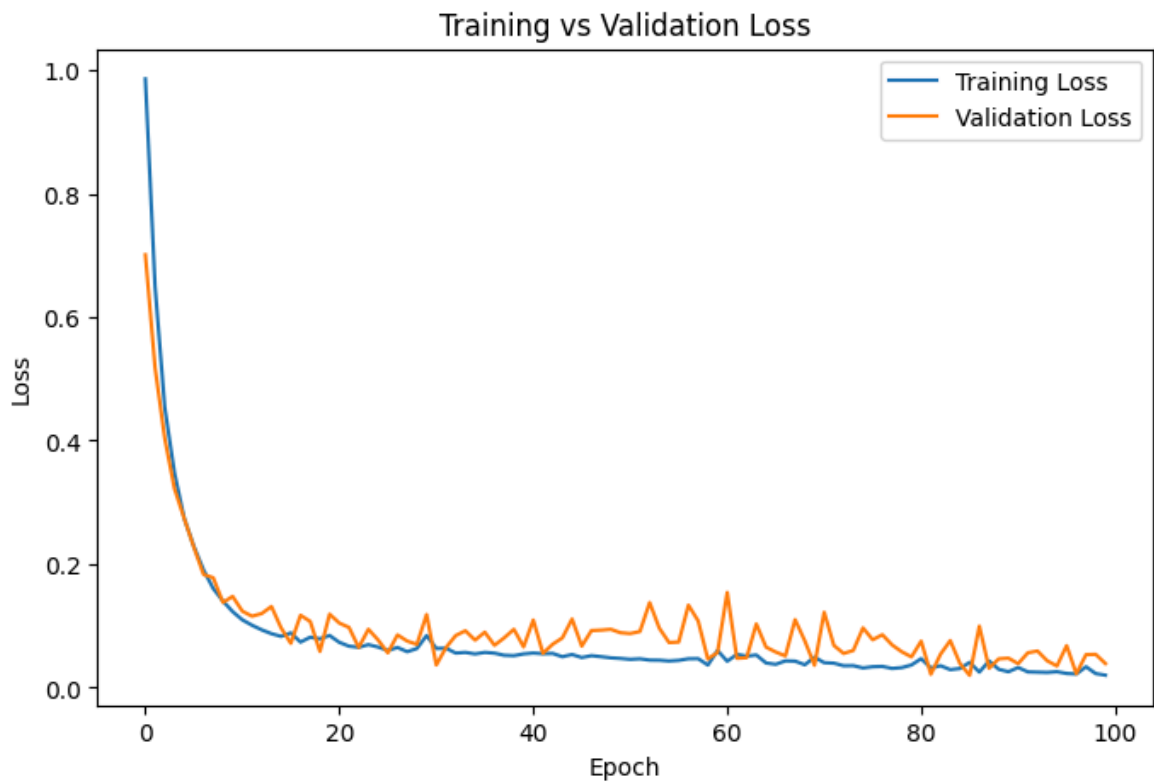
plt.xlabel("Epoch")

plt.ylabel("Loss")

plt.title("Training vs Validation Loss")

plt.legend()

plt.show()
```



```
In [18]: experiments = [
    ("sigmoid", 0.01),
    ("sigmoid", 0.1),
    ("relu", 0.01),
    ("relu", 0.1)
]

results = []

for activation, lr in experiments:

    model = Sequential([
        Dense(16, activation=activation, input_shape=(4,)),
        Dense(8, activation=activation),
        Dense(3, activation='softmax')
    ])

    model.compile(
        optimizer=Adam(learning_rate=lr),
```



```

        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )

    model.fit(
        X_train,
        y_train,
        epochs=50,
        batch_size=16,
        verbose=0
    )

    loss, acc = model.evaluate(X_test, y_test, verbose=0)

    results.append([activation, lr, acc])

results_df = pd.DataFrame(
    results,
    columns=["Activation Function", "Learning Rate", "Accuracy"]
)

results_df

```

c:\Users\msi\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\core\dense.py:95: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```

super().__init__(activity_regularizer=activity_regularizer, **kwargs)

```

Out[18]:

	Activation Function	Learning Rate	Accuracy
--	---------------------	---------------	----------

0	sigmoid	0.01	0.966667
1	sigmoid	0.10	1.000000
2	relu	0.01	1.000000
3	relu	0.10	1.000000

In [19]:

```

best = results_df.loc[results_df["Accuracy"].idxmax()]

print(best)

```

```

Activation Function    sigmoid
Learning Rate         0.1
Accuracy              1.0
Name: 1, dtype: object

```